

Scalable Data Aggregation for Robotics: A Multiagent Knowledge Graph System

Christopher Allred*, Jacob, Chandler Justice, and All the cool people too, and Mario Harper*[‡]

*Department of Computer Science, Utah State University

[‡]Computational & Information Sciences Directorate, Army Research Lab (ARL)

Abstract—This paper presents a multiagent system designed to aggregate robot data into a structured knowledge graph. The proposed system utilizes multiple autonomous agents, each tasked with collecting, processing, and integrating data from various robotic sources. The knowledge graph serves as a comprehensive repository, enabling efficient data retrieval, enhanced interoperability, and improved decision-making for robotic applications. By leveraging the multiagent architecture, the system ensures scalability, flexibility, and robustness in handling diverse and dynamic data inputs. Experimental results demonstrate the effectiveness of the system in constructing a coherent knowledge graph from heterogeneous robot datasets, paving the way for advanced robotics research and development.

Index Terms—Reinforcement Learning, Dynamic Motions, Legged Robots, Simulation

I. INTRODUCTION

This section provides an overview of the need for structured data aggregation in robotics. It discusses the importance of a knowledge graph in enhancing data retrieval, interoperability, and decision-making. The section introduces the multiagent system approach used in this paper.

II. SYSTEM DESIGN

The design of the proposed multiagent system is detailed in this section. It describes the multiagent architecture and the roles and responsibilities of autonomous agents in the system, including data collection, processing, and integration. The advantages

of using a multiagent system, such as scalability, flexibility, and robustness, are also discussed.

III. KNOWLEDGE GRAPH CONSTRUCTION

This section explains how the knowledge graph is constructed from various robotic data sources. It covers the structure and format of the knowledge graph and discusses the benefits it provides for robotics applications.

IV. EXPERIMENTAL SETUP AND RESULTS

The experimental setup and the datasets used to evaluate the system's effectiveness are described here. The section outlines the metrics and methods used for evaluation and presents the experimental results that demonstrate the system's ability to handle heterogeneous data and construct a coherent knowledge graph.

V. DISCUSSION

An analysis of the experimental results is provided in this section. It discusses the implications of the findings for robotics research and development and compares the proposed method with existing data aggregation techniques.

VI. CONCLUSION

The conclusion summarizes the key contributions of the paper and suggests potential future work to enhance the system or explore new applications.

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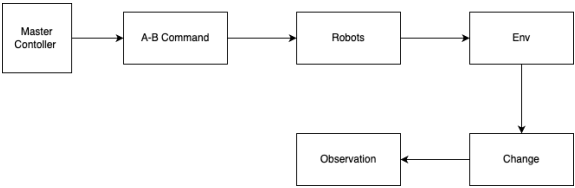


Fig. 1. Robot Knowledge Graph System